

Prototyping Autonomous Vehicle Lane Detection for Snow in VR

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ABSTRACT

Autonomous vehicles (AVs) are gaining momentum, and features such as autopilot are becoming widespread among consumer vehicles. AVs track the road to drive correctly, however, when they fail, the driver must take over. Lane detection is a must-have feature for AVs, which has numerous investigations. Nevertheless, most have gaps in extreme weather, such as snowy winters. The lack of snow datasets adds to this, as most include mild scenarios. This paper presents the prototype of a virtual reality digital twin that will allow training lane detection using synthetic data that would otherwise be difficult to recreate in real life.

Index Terms: Human-centered computing—Human computer interaction—Empirical studies in HCI—; Human-centered computing—Human computer interaction—Interaction paradigms—Virtual reality

1 INTRODUCTION

Lane detection is an important feature of autonomous vehicles that ensures safety by keeping the vehicle on the road despite any given conditions [14]. Common approaches to lane detection are based on computer vision techniques and machine learning solutions, such as Laneformer [3], which is an approach based on the transformer encoder-decoder architecture that explicitly models global semantic contexts by considering the results of lane, pedestrian and vehicle detection from a single unified network perspective. Similarly, in the study by Zou et al. [15], a deep hybrid architecture for lane recognition is developed, which combines a Convolutional Neural Network (CNN) for feature extraction, and a Recurrent Neural Network (RNN) for lane prediction. With the Self-Attention Distillation (SAD) approach, Hou et al. [4] aim to complement segmentation-based supervised learning by allowing the network to use the resulting attention maps of its own layers as distillation targets for its lower layers. Such approaches require access to data to properly train machine learning algorithms and ensure adequate detection and tracking [13]. However, existing methods are not suitable for extreme conditions, such as harsh winters with whiteouts that can make roads difficult to see [2], in addition to a lack of available snow road open databases [11]. In addition to the lack of adequate datasets, lane detection testing requires drivers behind the wheel to take control in cases where detection fails. Due to such limitations, an alternative to collecting suitable road data is to drive in harsh winter conditions. However, this poses several health and safety risks for drivers and equipment [7].

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Figure 1: Lane detection, Université du Québec à Trois-Rivières.

A solution to the limitations of real-world data collection has seen the use of simulation and synthetic data [6]. For example, NVIDIA Omniverse™, an open and extensible computational platform that enables the development of the NVIDIA DRIVE Sim™ [9], a multi-sensor simulation platform, where synthetic data, as well as digital twins, can be accessed to test and validate AVs. Although cloud-based services, such as NVIDIA Omniverse™, provide a suitable platform to take advantage of simulation, VR, and digital twins, costs and flexibility can introduce entry barriers. This article focuses on prototyping a VR digital twin for autonomous vehicle lane detection in snow conditions to capture additional data in a cost-effective manner while keeping test drivers safe. The approach presented in this paper builds on previous work that compares the accuracy of lane detection on a road with real snow and artificial snow generated using Photoshop effects.

2 MATERIAL AND METHODS

Our current method for finding lanes uses Lane Shape Prediction with Transformers (LSTR) [8], which is a parameter and transformer-based network trained from start to finish with a Hungarian fitting loss. This network learns from the global context and lane structures, and outputs the parameters of a lane shape model derived from the camera pose, and the road geometry. LSTR fine-tuning includes image enhancement techniques such as scaling, cropping, rotating, and flipping, which, complemented with the addition of artificial snow, provided the following prediction accuracy results of 68.76% with real road images (configuration a) (Fig. 1a), 77.38% with and without snow on real road images (configuration b) (Fig. 1b), and 89.08% with and without real and artificial snow on the real road (configuration c) (Fig. 1c). However, such an approach requires real-world road images, which may not be possible in certain winter conditions due to safety hazards.

The digital twin focused on recreating the real road used in previous work (Fig. 1), using a combination of Google Maps, Google Earth, and OpenStreetMap to obtain the layout and base 3D models to recreate the roads. This process required significant 3D authoring to ensure the 3D models adequately represent the road and relevant surroundings. Once the roads were completed, the 3D models were imported to the Unity game engine 2021.3.4f1. Unity was chosen due to its extensibility and compatibility with VR headsets and racing simulation wheels. Next, snow was added using Unity's particle system using texture changes to depict snow accumulation.

After finalizing the 3D model, a replica of the vehicle used for

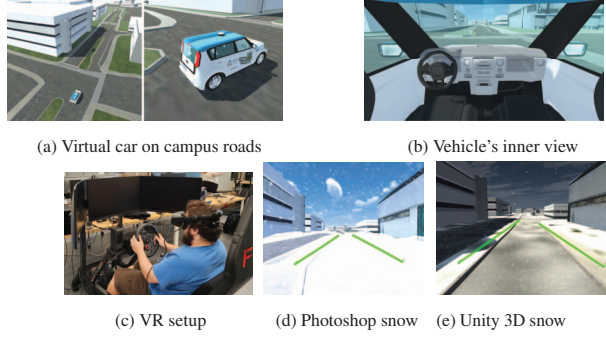


Figure 2: VR lane detection digital twin.

Table 1: Network performance on a dataset comparison

	Training a	Training b	Training c
Real road	52.9%	77.1%	64.3%
Photoshop snow	34.3%	36.4%	40%
Unity snow	37.1%	44.3%	31.4%

road tests was added Fig. 2b [1, 5, 10]. The virtual vehicle was configured by defining its mass, in addition to colliders and rigid bodies for the chassis and wheels to ensure proper physics behavior and steering range, including suspension spring and damper settings, motor force, and breaking force. The Logitech G29 steering wheel and shifter were integrated into the scenario for added realism Fig. 2c.

To further enhance immersion and presence, the Next Level Racing F-GT simulator cockpit with a monitor stand was used to attach the steering wheel and shifter in addition to VR integration using the HP Reverb G2 Omnicept Edition that offers non-invasive eye tracking and heart rate monitoring. Integration of the HP Omnicept VR headset required the import of the Windows Mixed Reality Toolkit (MRTK).

Lane detection accuracy was calculated using TuSimple metrics [12] to evaluate and compare the performance of the networks trained with the three aforementioned data configurations using the following three data sets: i) real road, ii) digital twin with artificial snow generated using Photoshop effects, Fig. 2d, and iii) digital twin with snow simulated in Unity, Fig. 2e.

3 PRELIMINARY RESULTS

In general, a decrease in network performance was observed compared to that obtained with real road images. Table 1 shows the performance of the networks on a set of data corresponding to the same road segment. For the network trained with the first configuration (a), a decrease in network performance was observed with respect to the real images of 18.6% and 15.7% for the digital twin data with Photoshop snow and with snow simulated in Unity, respectively. For the network trained with configuration (b), this decrease was 40.7% and 32.9% for images with Photoshop snow and snow particles, respectively. Finally, for the network trained with the last configuration (c), for images with Photoshop snow, the performance decreased 24.3%, and for images with Unity snow 32.9%.

4 CONCLUSION

This paper presented the prototyping of a VR digital twin for AV lane detection in snowy conditions. It offers a solution to overcome the limitations of data collection for neural network training, associated with driver safety, time, and cost. Likewise, it serves as a platform for user experience evaluation and monitoring of driver data when interacting with an AV, and in cases where lane detection fails.

Future work will focus on implementing within the digital twin the lane-detection neural network trained with all conditions to monitor and understand how test drivers respond to lane-detection failures that force them to take control of the vehicle. Additionally, while the use of the racing chair, steering wheel, and VR headset increases immersion, this will be further investigated to understand how this impacts responses in emergency lane detection failure.

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